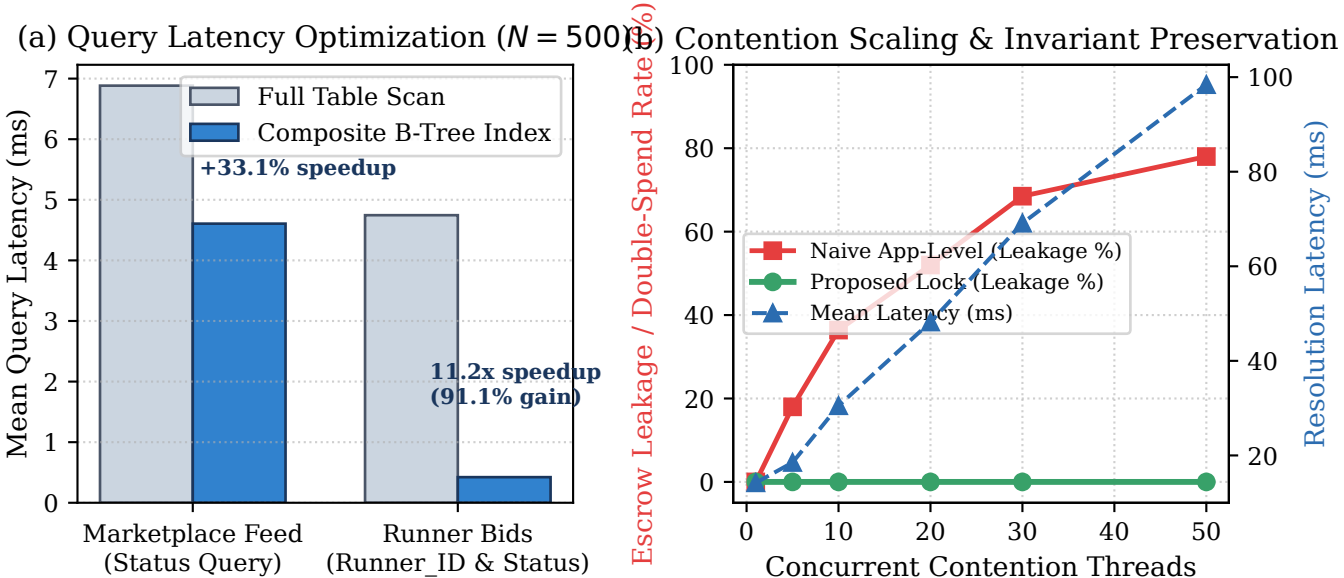
(a) Query Latency Optimization ($N = 500$)
Mean Query Latency (ms)
Full Table Scan
Composite B-Tree Index
+33.1% speedup
11.2x speedup
(91.1% gain)
Marketplace Feed
(Status Query)
Runner Bids
(Runner_ID & Status)
(b) Contention Scaling & Invariant Preservation
Escrow Leakage / Double-Spend Rate (%)
Resolution Latency (ms)
Naive App-Level (Leakage %)
Proposed Lock (Leakage %)
Mean Latency (ms)
Concurrent Contention Threads